

Executive Scenario Report

Predicted Load	Uncertainty	Financial Exposure	Grid Status
120.3 MWh	+/- 7.0 MWh	\$16,123	Elevated

Strategic Overview

This is a short-term MICRO operational forecast predicting an energy value of 120.32 with an uncertainty of 6.99. The ensemble model, heavily weighted towards CatBoost, indicates a slight decrease from the 24-hour rolling mean but an increase from recent lagged values. The forecast suggests a stable operational period with minor fluctuations.

Critical Risk Factors

- High reliance on the CatBoost model (65% weight) introduces a single point of failure if its underlying assumptions or data quality degrade.
- Negative contributions from recent lagged values (lag_12h, lag_24h) suggest a potential underlying downward pressure that might be underestimated by the current prediction.
- Unforeseen operational disruptions or rapid changes in unmodeled external factors could lead to significant deviations from this short-term forecast.

Actionable Recommendations

- Continuously monitor the performance of the CatBoost model and periodically re-evaluate ensemble weights to ensure optimal predictive accuracy.
- Investigate the drivers behind the negative contributions of recent lagged values to better understand short-term demand dynamics and potential shifts.
- Implement real-time anomaly detection and integrate additional high-frequency operational data streams to enhance responsiveness to sudden changes.

Input Feature Context

Hour:	12
Day Of Week:	2
Month:	1
Is Weekend:	0
Lag 24H:	120
Lag 12H:	118

Roll Mean 24H:	125
Roll Std 24H:	4